

SANCHU SRIJAN

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PROFESSIONAL SUMMARY

Data Science and Machine Learning enthusiast with project experience in Deep Reinforcement Learning, autonomous driving simulation, and control logic design. Specialized in training reactive agent models and exporting standards-compliant co-simulation Functional Mock-up Units (FMUs).

EDUCATION

B.Tech / M.Sc in Data Science / Computer Science

Specialization: Machine Learning & Intelligent Vehicles | Expected Graduation: 2027

TECHNICAL SKILLS

- * Programming: Python (3.11+), SQL, C++
- * Reinforcement Learning: PyTorch, Gymnasium, Stable-Baselines3 (PPO, DQN, DDPG)
- * Simulation Frameworks: MetaDrive driving simulator, OpenPASS integration
- * Automotive Standards: ASAM OpenSCENARIO XML format, Functional Mock-up Interface FMI 2.0 co-simulation
- * Exporters/Tools: pythonfmu, fmpy, Git, Markdown

ACADEMIC PROJECTS

Reinforcement Learning-Based Human Driver Behaviour Simulation (UST Drive)

- * Wrapped MetaDrive simulator in a custom Gymnasium environment featuring 259 lidar-state observations.
- * Authored multi-objective reward functions balancing safety cushion, headway, and jerk acceleration metrics.
- * Trained actor-critic models for Cautious and Aggressive driver profiles using SB3 PPO.
- * Exported policy models to FMI 2.0 co-simulation slaves (FMU format) and validated FMI variables with FMPy.
- * Modeled five test scenarios (S1-S5) utilizing ASAM OpenSCENARIO XML structures.